

PROCESS FLOWSHEET

Development and Characterization of Bio-Composites from Waste Aquatic Biomass for Sustainable Insulation | Dept. of Chemical Engineering, NIT Srinagar

Raw Material Collection

S1

Water hyacinth (*Eichhornia crassipes*) and **water lily** (*Nymphaea* spp.) harvested manually from **Dal Lake, Kashmir**. Petioles, stems, and leaves collected, washed with water to remove mud, debris, and biological contaminants.



Sun Drying

S2

Washed biomass spread on clean surfaces and **sun-dried for 5–7 days** under natural sunlight. Moisture removal reduces microbial activity. Observable colour change from green → tan/brown indicates moisture loss.



Oven Drying

S3

Sun-dried biomass transferred to a **drying oven at 105°C for 24 hours** to remove residual moisture and standardize moisture content prior to grinding. (ASTM D4442)



Grinding

S4

Oven-dried biomass ground using a **mixer grinder** (Organic Chemistry Lab, under Prof. Tabassum Ara, Dept. of Chemistry, NIT Srinagar) to reduce particle size for composite fabrication.

Sieving & Particle Size

Classification

S5

Ground biomass sieved using a **Sethi Standard Test Sieve shaker** (ASTM E11) to classify particles:

- **Coarse fraction:** >3 mm (< ASTM No. 7 mesh)
- **Fine fraction:** 1.0–1.5 mm (ASTM No. 12–18 mesh)



Binder Preparation (Parallel Step)

S6

Corn starch gelatinized on a **Tarsons Digital Spinot hot plate with magnetic stirrer** — temperature: 70°C, stirring: 150 RPM. Starch dissolved in water and heated until gel forms. Used as natural binder for S3 and S4.



Composite Mixing & Mould Filling

S7

Biomass particles mixed with binder/water in defined ratios (see sample variants below) and packed into **wooden moulds** of defined dimensions.

Hydraulic Pressing

S8

Filled moulds compressed using a **Hydraulic Press Model M-30** (Technosearch Instruments) to consolidate composite mixture, reduce porosity, and improve inter-particle bonding.



↓ **DEMOULDING → 4 COMPOSITE SAMPLES FABRICATED** ↓



SAMPLE 1 (S1)

Coarse particles (>3 mm)
Biomass + 15% water (no binder)
5×5×0.6 cm

⚠ **Fractured on demoulding**

SAMPLE 2 (S2)

Fine particles (1–1.5 mm)
Biomass + 15% water (no binder)
5×5×0.6 cm

✓ **Intact panel**

SAMPLE 3 (S3)

Fine particles (1–1.5 mm)
Biomass:Starch = 90:10
5×5×0.5 cm

✓ **Improved surface**

SAMPLE 4 (S4)

Fine particles (1–1.5 mm)
Biomass:Starch = 70:30
4.8×4.8×0.3 cm

✓ **Compact & dense**

S9

Characterization & Testing

All four samples subjected to the following tests:

Moisture Content

ASTM D4442 | Range: 9.94–41.18%

✓ **Completed**

Bulk Density

ASTM D1037 | Range: 0.868–1.087 g/cm³

✓ **Completed**

Water Absorption (2-hr soak)

ASTM D570 | Range: 280–658%

✓ **Completed**

Mechanical Strength

ASTM D695 (compressive) / ASTM D790 (flexural)

⌚ **Pending**

Flammability

ASTM D635 / UL-94 / ISO 11925-2

⌚ **Pending**

Thermal Conductivity

ASTM C518 / ASTM C177 — Target: 0.04–0.07 W/m·K

⌚ **Pending**